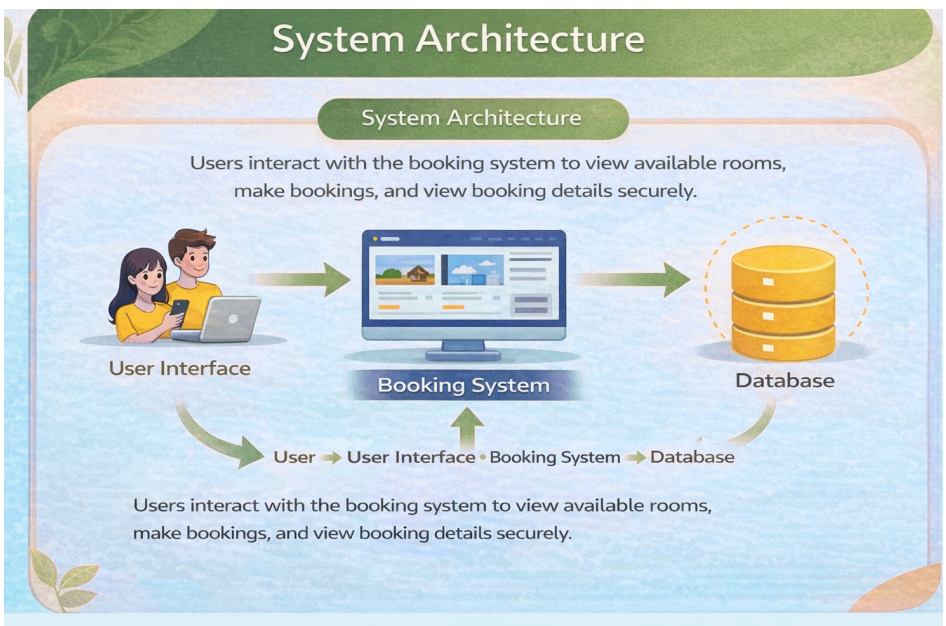
Hotel Room Booking Application Project

Project Based Learning - DSA

Abstract

The Hotel Room Booking System is developed using Java to demonstrate the implementation of fundamental Data Structures and Algorithms. The system allows users to view rooms, bookrooms and display booking details. Various DSA concepts such as Searching, Sorting , Linked Lists, Stacks, Queues , Hashing and Trees are applied to manage and process data efficiently.



CO-Wise Implementation

CO 1 Algorithm Analysis & Searching/Sorting

- Implemented searching algorithms (Linear Search, Binary Search) to locate data efficiently.
- Applied sorting algorithms (Bubble Sort, Selection Sort, Insertion Sort) to arrange data in order.
- Demonstrated understanding of algorithm efficiency for handling booking-related data.

CO 2 Abstract Data Types (Arrays / Lists)

- Implemented Linked List using Node and LinkedListExample classes.
- Performed ADT operations such as insertion and traversal of nodes.
- Organized structured data using dynamic list structures.

CO 3 Stack / Queue Data Structure

- Implemented Stack using array (push, pop) operations.
- Implemented Queue using circular array (enqueue, dequeue).
- Demonstrated real-world workflow processing using stack and queue concepts.

CO 4 Hashing & Java Collections

- Used Java Collection Framework (ArrayList) to store and manage booking records.
- Managed dynamic hotel booking data efficiently.
- Applied collection operations for fast data insertion and retrieval.

Conclusion & Future Scope

This project demonstrates the implementation of Data Structures and Algorithms using Java to develop a simple hotel booking system. It uses concepts such as searching, sorting, stack, queue, linked list, and collections to manage booking data efficiently. In the future, the system can be enhanced by adding database integration for permanent data storage, online payment options, real-time room availability checking, and a web or mobile interface to improve usability and functionality.

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